

Proof. **(1) Norm expressed by the inner product.** By definition,

$$\|v\|^2 = \langle v, v \rangle, \quad v \in V. \quad (1)$$

(2) Square the inequality. Since norms are non-negative, $\|x + y\| \leq \|x\| + \|y\|$ is equivalent to

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (2)$$

(3) Expand the left-hand side. Using bilinearity (or sesquilinearity) of the inner product,

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned}$$

(4) Expand the right-hand side and cancel.

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2.$$

Substituting this and (??) into (??) gives

$$\|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2,$$

hence, after cancelling the equal terms $\|x\|^2$ and $\|y\|^2$,

$$2 \operatorname{Re} \langle x, y \rangle \leq 2\|x\|\|y\| \iff \operatorname{Re} \langle x, y \rangle \leq \|x\|\|y\|. \quad (4)$$

(5) Apply Cauchy–Schwarz. The Cauchy–Schwarz inequality states

$$|\langle x, y \rangle| \leq \|x\|\|y\|. \quad (\text{CS})$$

Since $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$, (??) implies (??).

(6) Conclude the squared inequality. Insert the bound $\operatorname{Re} \langle x, y \rangle \leq \|x\|\|y\|$ into (??):

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2.$$

Both sides are non-negative; taking the principal square root preserves the inequality:

$$\|x + y\| \leq \|x\| + \|y\|.$$

(7) Degenerate cases. If $x = 0$ or $y = 0$ (or both), the equality $\|x + y\| = \|x\| + \|y\|$ holds trivially.

(8) Equality condition. Equality in the triangle inequality occurs precisely when equality holds in the Cauchy–Schwarz step (??). Equality in Cauchy–Schwarz is equivalent to linear dependence with a non-negative proportionality factor, i.e.

$$\exists \lambda \geq 0 \text{ such that } y = \lambda x \quad (\text{or } x = 0 \text{ or } y = 0).$$

In this case the vectors point in the same direction and $\|x + y\| = \|x\| + \|y\|$.

Thus the triangle inequality holds for all vectors in an inner-product space, with equality exactly in the described linear-dependence situation. $\square$ $\square$